WEEK 3 · LECTURE L1 · UNIT I

Data Literacy and Statistical Thinking

How to read an HR number without being fooled: populations and samples, rates and counts, base rates, and correlation versus causation.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Distinguish population, sample and self-selection, and explain why selection breaks HR data.
- Choose correctly between counts and rates, and percent and percentage points.
- Explain base-rate neglect, Simpson's paradox, survivorship and regression to the mean with HR examples.
- Evaluate a causal claim using the counterfactual view and the Bradford Hill considerations.

COURSE OUTCOMES

CLO 1 (data literacy) and CLO 3 (evaluating people analytics).

DATA LITERACY, ONE OF THREE

Who is actually in this number?

Population

Everyone you mean to describe. HR files rarely contain it: leavers drop out, contractors are excluded, new joiners are not yet loaded.

Sample

The subset you actually have. It supports a claim about the population only if you know how it was selected.

Self-selection

The dangerous case. Survey respondents chose to answer; app users chose to log in. Those missing are missing for a reason linked to what you measure.

EXAM LENS

Ask who is missing before you ask what the number says. Selection, not sampling error, is what usually breaks HR data.

DATA LITERACY, TWO OF THREE

A count tells you size; a rate tells you risk

Claim	What it is	Where it misleads
"Store operations lost the most people"	A count	The largest department almost always loses the most people
"Store operations has the highest attrition rate"	A rate: leavers over a base	Only as good as the base: which headcount, when, including whom?
"Attrition rose from 31% to 38%"	A rate over time	A falling denominator raises a rate without anyone extra leaving
"Team X has 17% attrition"	A small-base rate	One departure from a team of six is 17%; small bases make rankings unstable

DATA LITERACY, THREE OF THREE

An average can hide the whole story

Means hide spreads

An organization-wide mean digital literacy score of about 3 out of 5 may describe almost nobody if metro staff cluster high and small-town staff cluster low.

Split before you conclude

Segment by location, grade, device and contract type before drawing any conclusion from an average.

Report the distribution

Share scoring low and share scoring high often say more than the mean.

THREE HR DATA TRAPS, PLUS BASE RATES

Base rates, survivorship, Simpson, regression

The base-rate trap

A flight-risk flag catches 9 of 10 leavers but also 9 of 90 stayers. If 10 of 100 people leave, half of everyone flagged was never going to leave.

Survivorship

Exit interviews hear only from leavers; surveys only from those who answer. The silent group is often the one you most need.

Simpson's paradox

A gap can reverse when data are split: a firm-wide pattern may vanish or flip inside every department.

Regression to the mean

The worst unit this quarter usually improves next quarter anyway. Do not credit the intervention automatically.

BEFORE CLAIMING CAUSATION

Four things a correlation can be

Explanation	Meaning	HR example
A causes B	The claim you want; needs a mechanism and ideally variation you did not choose	Training raises performance
B causes A	Reverse causation	Engaged staff volunteer for training
C causes both	A confounder	Grade drives device access, literacy and app usage at once
Selection made it	The pattern lives in how the file was built	Only active users appear in usage logs

EXAM LENS

A strong answer names and rules out the three rival explanations before accepting a causal claim.

PRECISION THAT EARNS MARKS

Percentage points, full scale and the right denominator

Points, not percent

A rate rising from 20% to 30% rises by 10 percentage points, which is a 50 percent increase. Say which you mean.

Sample or full scale

If a file holds a one-in-eight sample, a Board figure must be scaled up, and the scaling stated.

The right base

Divide by the people actually in the table, not by the size of a different file.

Read every number aloud with its unit and its base: "a 10 percentage point rise, among 300 workers".

THEORY IN DEPTH · TVERSKY AND KAHNEMAN (1973)

We neglect base rates

THE CORE IDEA

People judge probability by how easily examples come to mind and how well a case fits a story, neglecting how common the outcome is.

HOW IT WORKS

A vivid profile overrides the base rate. The fix is to start from the base rate, then adjust.

LIMITS AND CRITIQUE

Base rates are used much better when problems are framed as natural frequencies (Gigerenzer and Hoffrage, 1995).

IN PRACTICE

A flight-risk flag must be read against the organization's actual attrition rate, or most flagged people will be false alarms.

EXAM LENS

Show the arithmetic: a short natural-frequency example (out of 100 people) is more convincing than a percentage.

THEORY IN DEPTH · BICKEL, HAMMEL AND O'CONNELL (1975)

Simpson's paradox: the aggregate can mislead

THE CORE IDEA

University of California, Berkeley graduate admissions appeared to favor men overall.

HOW IT WORKS

Department by department, women were admitted at similar or slightly higher rates; women had applied more to competitive departments.

LIMITS AND CRITIQUE

The right level of analysis depends on the causal question, not on the data alone.

IN PRACTICE

An organization-wide gender gap in job offers must be checked within each role and location before drawing a conclusion.

EXAM LENS

Use this case whenever a question asks you to evaluate an aggregate HR statistic.

THEORY IN DEPTH · RUBIN (1974); HOLLAND (1986)

Causation means comparing with the counterfactual

THE CORE IDEA

The effect of training on a person is the difference between their outcome with and without it.

HOW IT WORKS

We see only one outcome, so we compare groups; random assignment makes the groups comparable.

LIMITS AND CRITIQUE

Without randomization, hidden differences between groups can create a false effect.

IN PRACTICE

Employees who chose a course may differ from those who did not; compare like with like before crediting the course.

EXAM LENS

Holland's "fundamental problem of causal inference" is a precise phrase worth using in critical answers.

THEORY IN DEPTH · BRADFORD HILL (1965)

Nine considerations for judging a causal claim

THE CORE IDEA

When experiments are impossible, weigh several kinds of evidence for causation.

HOW IT WORKS

Strength, consistency, specificity, temporality, biological gradient (dose-response), plausibility, coherence, experiment and analogy.

LIMITS AND CRITIQUE

Only temporality is strictly necessary; the rest are guides, not a checklist to add up.

IN PRACTICE

Does higher engagement come before lower attrition, and do more engaged teams leave less? Test temporality first.

EXAM LENS

Hill called these "viewpoints", not criteria. Applying two or three carefully beats listing all nine.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"HR data rarely lie, but they often mislead." Discuss with reference to selection, base rates and Simpson's paradox. **20 marks**

A strong answer: explains each trap with an HR example; shows the base-rate arithmetic; uses the Berkeley case; concludes with practical safeguards.

A firm finds that employees who complete training have lower attrition and concludes training reduces attrition. Critically evaluate this conclusion. **20 marks**

A strong answer: lays out reverse causation, confounding and selection; applies the counterfactual view and Bradford Hill; proposes a design that could test the claim.

Distinguish a percentage change from a change in percentage points, with an example. **10 marks**

A strong answer: gives a correct worked example and explains why the distinction matters in HR reporting.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **Every rate has a numerator and a denominator**
   Check both before comparing.

2. **Means hide spreads**
   Split by segment before you conclude.

3. **Correlation is a clue, not a cause**
   Rule out reverse causation, confounding and selection.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Kavanagh, M. J., & Johnson, R. D. (2018), Ch 2 (data, information and knowledge; warehouses, BI and data mining) and Ch 14.
- Angrave, D., et al. (2016). HR and analytics: Why HR is set to fail the big data challenge. *Human Resource Management Journal, 26*(1), 1–11.

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Bickel, P. J., Hammel, E. A., & O'connell, J. W. (1975). Sex bias in graduate admissions: Data from Berkeley. *Science, 187*(4175), 398–404. https://doi.org/10.1126/science.187.4175.398

Gigerenzer, G., & Hoffrage, U. (1995). How to improve Bayesian reasoning without instruction: Frequency formats. *Psychological Review, 102*(4), 684–704. https://doi.org/10.1037/0033-295X.102.4.684

Hill, A. B. (1965). The environment and disease: Association or causation? *Proceedings of the Royal Society of Medicine, 58*(5), 295–300. https://doi.org/10.1177/003591576505800503

Holland, P. W. (1986). Statistics and causal inference. *Journal of the American Statistical Association, 81*(396), 945–960. https://doi.org/10.1080/01621459.1986.10478354

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology, 66*(5), 688–701. https://doi.org/10.1037/h0037350

Tversky, A., & Kahneman, D. (1973). Availability: A heuristic for judging frequency and probability. *Cognitive Psychology, 5*(2), 207–232. https://doi.org/10.1016/0010-0285(73)90033-9